00-S-101B
June 25, 1973
SUPERSEDING
Fed. Spec. 00-S-101A
May 13, 1965

FEDERAL SPECIFICATION

SANDER, DISK AND OSCILLATING PAD, PNEUMATIC, PORTABLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers portable, pneumatic, disk and pad sanders, used for sanding and polishing on flat and curved surfaces.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the title of the specification, but are intended to cover those which are most generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and styles. Portable pneumatic sanders shall be of the following types and styles, as specified (see 6.2):

Type I - Sander, oscillating pad.
  Style 1 - Orbital motion.
  Style 2 - Reciprocating motion.

Type II - Sander, disk.
  Style 1 - Rotary motion, vertical drive.
  Style 2 - Rotary motion, right angle drive.
  Style 3 - Dual motion, vertical drive.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Handbook:

H28 - Screw-Thread Standard for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.)

American National Standards Institute (ANSI), Inc., Standard:

S5.1 - CAGI-PNEUROP Test Code for the Measurement of Sound from Pneumatic Equipment.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Preproduction model. When specified (see 6.2), the supplier shall furnish within the time frame specified (see 6.2), one sander of the type and style specified in the contract to prove, prior to starting production, that his production methods and choice of design detail will produce sanders that comply with the requirements of this specification. Examination and tests shall be those specified herein. Any changes or deviations from the preproduction model during production shall be subject to the approval of the contracting officer. Approval of the preproduction model by the contracting agency shall not relieve the supplier of his obligation to furnish sanders conforming to this specification.

3.2 Illustrations. The illustrations shown herein are descriptive, not restrictive, and are not intended to preclude the purchase of sanders otherwise conforming to this specification.

3.3 General construction. The sanders shall be of rugged construction so as to withstand, without failure, treatment encountered during actual service conditions. The sound pressure shall not exceed 90 dbA (see 4.8.3). Working parts subject to wear shall be of sufficient hardness to withstand the service required. Similar parts, from the same manufacturer, shall be interchangeable and of a good fit when assembled with their mating parts. The sanders shall be reliable and effective when operated at air line pressures between 80 and 100 pounds per square inch (p.s.i.). Sanders may be furnished which otherwise meet the requirements as stated but require less than 80 p.s.i. operating pressure. Each sander shall be equipped with a regulator, which may operate in conjunction with the throttle, to control the volume of air entering the motor.

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3.3.1 Materials. The materials used in the construction of all components shall be sound, of uniform quality and condition, and shall conform in composition, heat treatment, and suitability to the standard practices of manufacturer's producing tools of the types required in this specification. All parts shall be new and high grade commercial quality.

3.4 Motor. The sander shall be equipped with a rotary vane-type pneumatic motor consisting of one or more elements accurately balanced with the rotor shaft.

3.5 Throttle. The sander shall be equipped with a manually operated, lever type throttle. It shall be located so as to minimize possible accidental operation and shall automatically close the air inlet valve when actuating pressure is removed. When specified (see 6.2), a button type throttle conforming to these requirements shall be furnished.

3.6 Lubrication. Provisions shall be made for lubrication to be supplied to the rotor, cylinder, and other moving parts in the path of the air stream. The oil may be applied manually into the motor or supplied automatically through the air supply line. Other suitable means shall be provided for supplying lubrication to gears and other moving parts not in the air stream.

3.7 Air inlet connection. The sander shall have a female, 1/4 inch size, standard taper pipe thread, air inlet connection. The inlet shall be equipped with an air filter which shall retain solid particles in the air supply line. The filter shall be easy to remove, clean, and replace.

3.8 Bearings. All rotating parts of the sanders shall be provided with commercially available rolling contact bearings. The rotor shaft shall be provided with ball bearings. Needle and roller bearings, which are primarily designed for radial loads, shall not be used in a thrust position. If open type bearings are used, they shall be suitably housed and adequately sealed to prevent leakage of lubricant and entrance of dirt and dust.

3.9 Screw threads. All threaded parts shall conform to the applicable requirements of H28. When threaded fasteners are utilized to assemble adjacent housing components, through bolting shall be used in preference to tapped holes. Cap screws, if used, shall engage threads in ferrous housings or ferrous inserts in nonferrous housings by not less than a length equal to the pitch diameter of the fastener thread. Where ferrous inserts are not used in nonferrous housings, cap screws shall engage threads by not less than a length equal to 1-1/2 times the pitch diameter of the fastener thread. All threaded fasteners shall be secured with suitable locking devices.

3.10 Identification marking. Each sander shall be marked in a permanent and legible manner. The marking shall include the manufacturer's name or trademark and the model, size, or type designation so that the sander and the source of manufacture may be readily identified. The marking shall include the recommended operating air pressure when it is less than 80 p.s.i.

3.11 Carrying case and accessories. When specified (see 6.2), the sander shall be furnished in a durable metal or plastic carrying case. The case shall be provided with a handle, hinged cover, and at least one suitcase snap latch or equivalent means of holding the cover in place. The snap latch(es) or other cover holding means shall be of such design, strength, and constructions as not to open when the case and contents are being carried by the handle. The case shall be of such size as to hold the sander and its accessories. Accessories deemed necessary by the manufacturer for satisfactory operation and maintenance of the sander shall be furnished.

3.12 Finish and coatings. Sanders shall be finished and coated in accordance with the manufacturer's standard practice.

3.13 Accessories or extra equipment. When accessories or extra equipment are required, they shall be as specified (see 6.2).

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3.14 Instruction book. One copy of an instruction book consisting of the manufacturer's standard commercial instructions and parts list shall be furnished with each sander.

3.15 Workmanship. The sanders shall be free from all defects or imperfections which will adversely affect the general appearance, function, serviceability, or life expectancy.

3.16 Type I, oscillating pad sanders. The type I sanders shall conform to the applicable requirements of table I. The sanders shall be equipped with two effective, hand operated holders for securing the abrasive sheets to the pad and shall be equipped with a 1/4 inch minimum thick felt or soft rubber pad.

TABLE I. Type I, oscillating pad sanders

Style	Contact area sq. inches (min.)	No load speed cycles per minute (min.)	Air consumption at no load (c.f.m. max.)	Weight (lbs. max.)
1	16	5000	15	5
2	18	2000	15	5

3.16.1 Style 1, orbital motion. The type I, style 1 sanders shall be similar to figure 1. The sander shall approximately subscribe an orbit with a minimum diameter of 5/32 inch.

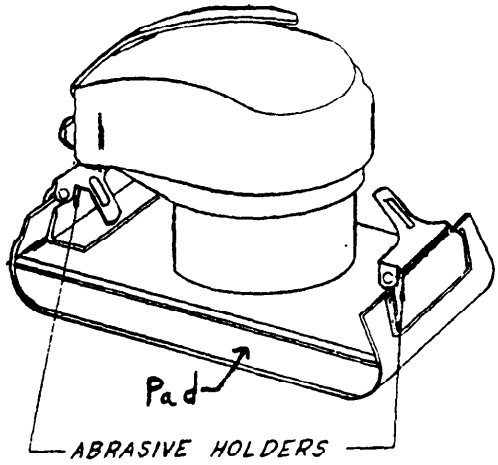


FIGURE 1. Type I, oscillating sander, style 1, orbital motion.

3.16.2 Style 2, reciprocating motion. The type I, style 2 sanders shall be similar to figure 2. The sanders shall operate in a straight line, reciprocating motion.

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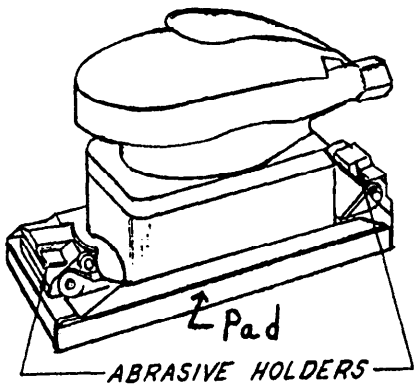


FIGURE 2. Type I, oscillating sander, style 2, reciprocating motion.

3.17 Type II, disk sanders.

3.17.1 The type II sanders shall conform to the applicable requirements of table II and shall be equipped with a felt, fiber, or soft rubber pad of the size specified in this table. The method of attaching the abrasive sheet shall be as recommended by the manufacturer and if special tools are required, they shall be furnished with the sander.

TABLE II. Type II, disk sanders

Style	Pad size diameter inches (min.)	No load speed (min.) (max.)	Air consumption at no load (c.f.m. max.)	Horse-power (min.)	Air consumption at minimum horsepower (c.f.m. max.)	Weight (lbs. max.)
1	4	10,000 12,000	25	0.35	60	6
2	7	5,000 7,000	35	0.60	70	9
3	4	8,000 10,000	20	0.35	40	5

3.17.2 Style 1, rotary motion, vertical drive. The type II, style 1 sanders shall be similar to figure 3. The sanding pad shall be driven directly from a vertical drive air motor.

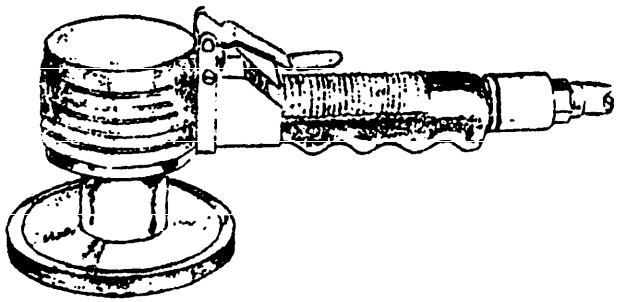


FIGURE 3. Type II, disk sander, style 1, rotary motion, vertical drive

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3.17.3 Style 2, rotary motion, right angle drive. The type II, style 2 sanders shall be similar to figure 4. The sanding pad shall be driven by the air motor at 90°. The sander shall be provided with an auxiliary side handle mounted at 90° to 120° from the control handle.

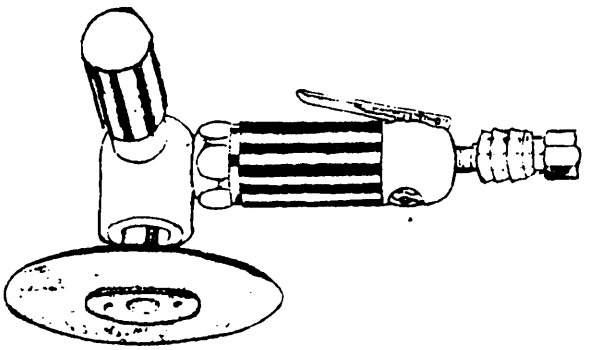


FIGURE 4. Type II, disk sander, style 2, rotary motion, right angle drive.

3.17.4 Style 3, dual motion, vertical drive. The type II, style 3 sanders shall be similar to figure 5 and be capable of developing two separate and distinct actions as viewed from the back of the pad as follows:

ACTION 1. A combination oscillating and rotating action wherein the pad face moves to the right 5,000 to 9,000 oscillations per minute and at the same time, rotates to the left (counterclockwise) 500 to 1,500 revolutions per minute.

ACTION 2. Eccentrically rotating action wherein the pad face rotates 5,000 to 9,000 revolutions per minute to the right (clockwise).

An integral shifting mechanism shall be incorporated which shall readily permit shifting from one action to the other without the use of special tools or equipment. When the sander is operating in either action, the vibrations developed shall not exceed 0.017 inch in either the horizontal or vertical plane (see 4.8.5).

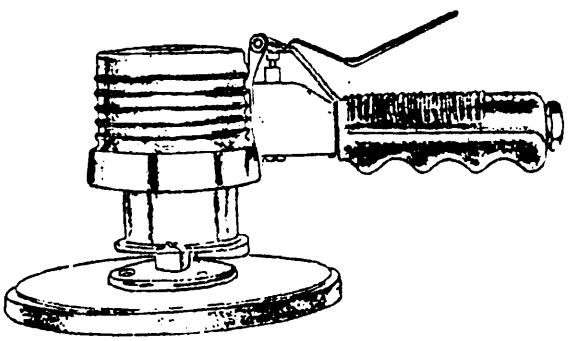


FIGURE 5. Type II, style 3, dual motion, vertical drive, sander.

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3.17.4.1 A locking device or selector shall be included to allow the operator to select either of the actions desired. A speed control device or selector shall be included to effectively control the speed of the sander from full to stop, without the need for tools, and shall be in addition to the throttle requirements of 3.5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.

(a) Preproduction inspection (see 4.3).
(b) Acceptance inspection (see 4.6).
(c) Inspection of preparation for delivery (see 4.7.2).

4.3 Preproduction inspection. Each type and style of sander shall be examined and tested as specified in 4.7 and 4.8. Presence of one or more defects shall be cause for rejection.

4.4 Lot. A lot for inspection purposes shall consist of all sanders of the same type and style submitted for inspection at the same time and place.

4.5 Sampling. Sampling for acceptance inspection shall be in accordance with inspection level S-2 of MIL-STD-105 with an Acceptable Quality Level (AQL) of 4.0 percent defective.

4.6 Acceptance inspection. Each sander shall be examined as specified in 4.7 and 4.7.1. Presence of one or more defects shall constitute a defective sander.

4.7 Examination. Each sander shall be examined for conformance to design and construction, dimensions, weight, materials, finish and coating, workmanship and marking requirements. The carrying case, accessories, and instruction book shall be examined for conformance to specified requirements.

4.7.1 Acceptance tests. Each sample shall be run at no load to determine integrity of working parts, noise, and vibration. Noise and vibration shall be subjectively evaluated to determine if operation is consistent with approved preproduction sample. Samples exhibiting excessive noise or vibration shall be tested in accordance with 4.8.3 and 4.8.5 to verify actual performance. Any sample failing to meet noise or vibration requirements shall constitute a defective.

4.7.2 Examination of preparation for delivery. An examination shall be made to determine compliance with preservation, packaging, packing, and marking in accordance with PPP-P-40.

4.8 Tests.

4.8.1 Test conditions. All performance tests shall be based on operation of the sanders at an air pressure of 90 p.s.i. at no load speed measured as close to the tool as practical. Sanders being tested which require less than 90 p.s.i. (see 3.3), shall be tested at the operating pressure specified by the manufacturer. Test sander shall be connected to the air supply by means of a 5/8 inch inside diameter hose containing a filter and a regulator. An air line lubricator may be used, if required.

4.8.2 Air consumption. The sander shall be tested to determine compliance with the requirements for air consumption which shall be measured with a liquid displacement meter or a flowmeter.

4.8.3 Sound pressure level test. The sound pressure level test shall be conducted in accordance with the applicable provisions of ANSI S5.1, except as modified herein. The sound pressure level shall not exceed 90 decibels on the A weighted scale at any microphone location measured at an unobstructed distance of one meter.

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4.8.3.1 Acoustical calibration. The entire instrumentation system shall be calibrated before each test series allowing a minimum instrument warm-up period of 5 minutes if transistorized and 1/2 hour if vacuum tube. Recalibration shall be performed if any instrument adjustment is necessary during progress of the test. All instrumentation (except microphone) and test personnel shall be as remotely located as practicable so that the measured data will be unaffected by a nonstationary observer.

4.8.3.2 Background sound level. The ambient noise level shall be no greater than 10 db less than the measurement taken with the tool running in all tool and microphone orientations. In addition, the sound produced by the energy absorbing device, used in the "on load" test, must be at least 10 dbA below the machine's own sound output.

4.8.4 Speed.

4.8.4.1 Type I. The type I sanders shall be tested to determine compliance with the speed requirements specified. A stroboscope or other suitable device, of sufficient capacity and accuracy, shall be used. For style 1, the orbital diameter shall be checked by whatever means necessary.

4.8.4.2 Type II. The type II sanders shall be tested to determine compliance with the speed requirements specified. The speed specified shall be spindle speed, with pad detached. The speed data shall be obtained with a tachometer of sufficient capacity and accuracy.

4.8.5 Dual motion tests. The type II, style 3 sander shall be tested to determine compliance with the requirements for double action and vibration (see 3.17.4). The sander shall be equipped with a 4-inch (bare) felt or canvas face sponge pad. Abrasive (sanding) discs shall not be used for this test. Sander to be operated at 90 p.s.i. or as specified by the manufacturer (see 3.3) with speed control valve and throttle fully open. The pad shall be placed on a smooth, flat testing surface of steel or plexiglass of a larger size than the pad diameter with a one-pound weight on top of the sander during test. The dual action and vibration values shall be verified by using test equipment of sufficient capacity and accuracy.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The preservation, packaging, and packing shall be in accordance with PPP-P-40 at the level specified (see 6.2) and the markings shall be in accordance with PPP-P-40.

6. NOTES

6.1 Intended use. The sanders covered by this specification are intended for use in shops for sanding operations on wood, metal, fiberglass, plexiglass, etc. In considering the service applications for sanders, they are normally used as follows:

Oscillating sanders - For final finish sanding.

Disk sanders, rotary motion - For heavy removal, metal and wood.

Disk sanders, dual motion - For heavy removal, or final finish sanding for prevention of swirl marks on wood, metal, plexiglass, etc.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and style required (see 1.2).
(c) Specify if preproduction model is required and include the date for submittal (see 3.1).
(d) Button type throttle, if required (see 3.5).
(e) Carrying case, if required (see 3.11).
(f) Accessories and extra equipment, if required (see 3.13).
(g) Level of preservation, packaging, and packing required (see 5.1).

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6.3 Maximum sound pressure levels have been set to insure against occupational noise-induced hearing losses. The tools covered by this specification are considered intermittent duty tools, used in short bursts of 2 minutes duration or less.

6.4 The sanders covered by this specification are in compliance with the requirements of the Occupational Safety and Health Administration (OSHA) as published prior to the date of this document.

6.5 Supersession data. This specification supersedes Interim Federal Specification 00-S-0095 dated March 21, 1956, and Military Specification MIL-S-9838 dated April 7, 1958, in part.

MILITARY INTERESTS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS
HEW-FDA

Preparing activity:

GSA-FSS

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